8.2 More Key Features of Quadratic Functions

Lesson Introduction

Benchmark	MA.912.AR.3.7 Given a table, equation, or written description of a quadratic function, graph that function and determine and interpret its key features.	Learning Targets	• I can identify key features of a quadratic given a graph, a table, and an equation. • I can graph a parabola given an equation or a table.

	Building On	Working Toward
Benchmark Coherence	N/A	N/A

Essential Questions	• How can the way a quadratic is represented provide information on key features? • How can knowing the key features from the equation or table help graph a parabola?
Lesson Narrative	In this lesson, students continue working with key features of quadratics. Students will be introduced to the following additional key features: end behavior, domain, and range. They will work with quadratics represented in graphs, tables, and equations in order to identify all the key features learned between this lesson and Lesson 1.

Component Summary	8.2.1 Warm-Up (~5 min.)	Arch analysis activity *(SE p. 10)*	8.2.3 Collaboration (10-15 min.)	Collaboration on determining the key features of a quadratic function *(SE p. 15)*
	8.2.2 Guided Instruction (20-25 min.)	Discover additional key features for quadratics *(SE p. 11)*	8.2.4 Your Turn! (10-15 min.)	Determine the key features of a quadratic independent practice *(SE p. 16)*
	8.2.5 Wrap-Up (~5 min.)	Key features self-reflection activity *(SE p. 16)*		

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms:</u> • End Behavior • Domain • Range <u>Additional Terms:</u> • Intercepts • Vertex • Axis of Symmetry	• Graphing Activity • (Optional) Chart Paper & Markers • (Optional) Colored Pencils or Highlighters	• If students have access to personal devices, consider using a share code for the class to engage digitally with your instruction during the lesson. If not, consider projecting the Desmos activity for students to engage as a whole group with the visuals and manipulatives.

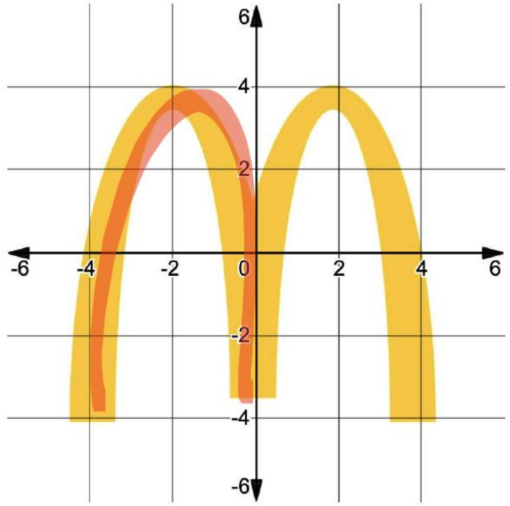
<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may not put the correct signs on infinity when notating the domain or range.
Students may get the domain and range values reversed.</td><td>Students have learned the other key features besides end behavior, domain, and range in Lesson 1. Activities 8.2.3 Collaboration and 8.2.4 Your Turn! have students looking for all key features (new and old). If time is short, consider having students only look for new ones from this lesson.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
8.2.3 Collaboration allows students to work with a partner to determine the key features of a quadratic given an equation and table of values. Students can think about the problem on their own, pair up with their partner, and share answers to come to a consensus.</td><td>MA.K12.MTR.2.1: Multiple Representations
Activities 8.2.3 Collaboration and 8.2.4 Your Turn! require students to find multiple key features of quadratics in various forms. Students should be making connections between the representations of a quadratic function. As stated in the clarifications, students should be given the opportunity to articulate connections between concept and representation while transitioning from pictorial to abstract.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Desmos Activity Builder
If resources permit, consider facilitating this lesson using the accompanying Desmos activity for Lesson 2. You can facilitate with the whole group using a projector or smart board while students watch or engage on their personal devices, or you can put students in small groups, pairs, or individually if resources permit. The digital interaction provides additional entry-points for students to work at their own pace while asynchronously engaging with the content and each other.</td></tr>
<tr><td colspan="3">Lesson Notes</td></tr>
</table>

8.2.1 Warm-Up: Henry’s Arches

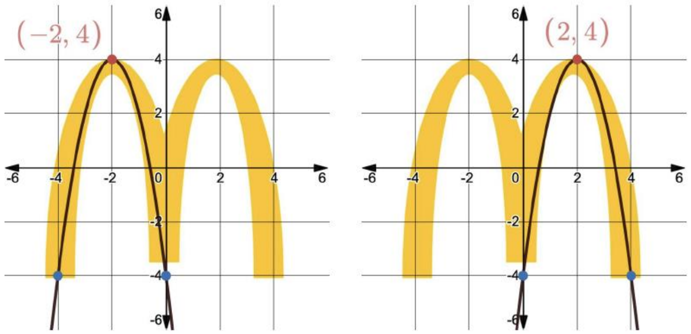
Component Overview: *Students apply the previous day’s learning to real-world arches. This activity is designed to be done independently. For this activity, release students to try these questions on their own without substantial support. Bring all students back together to review the answers as a whole group.*

8.2.1 Warm-Up: Henry’s Arches

1. Sketch a curve on the graph that fits one of the arches.



Henry used a computer program to draw curves that fit each of the arches. His curves are shown.



Consider displaying this image on a smart board, projector, or screen. If technology is unavailable, consider sketching the original image onto a piece of poster paper or on the whiteboard. When having students share out their answers, allow students to draw their original sketches on the master copy that is displayed for the class.

Elicit student responses that highlight attention to precision in graphing and key features.

- *What are the advantages of using a computer program to graph the arches instead of doing it by hand?*
- *Describe a situation where it would be more beneficial to do it by hand.*

8.2.1 Warm-Up: Henry’s Arches (continued)

2. What do you notice about the two curves?
   Answers will vary. Both curves appear to be the same size in height and width. Both curves look the same but share some of the same values.

The computer program also gives Henry the equations and a table of value for the curves.

Left Arch		Right Arch	
$y = -2(x-(-2))^2 + 4$ OR $y = -2(x+2)^2 + 4$		$y = -2(x-2)^2 + 4$	
x	y	x	y
-3	2	-2	-28
-2	4	-1	-14
-1	2	0	-4
0	-4	1	2
1	-14	2	4
2	-28	3	2

3. What is one thing you notice about the tables of values?
   Answers will vary. The y-values are flipped. They have the same y-value of the vertex, but opposite signs for the x-values.

4. Which key feature was the most useful in matching a quadratic to the arch?
   The vertex.

Allow multiple students to share out their answers to question 2. Consider polling the students to determine which students noticed something similar about the two curves and have them share their rationale for what was noticed.

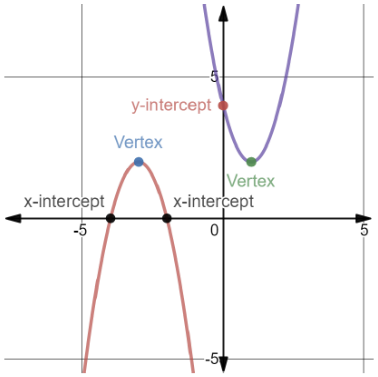
There may be more differences for question 3 than there were for question 2. Consider charting the things they noticed from the table of values in a central location, notating multiple presentations of the same noticing.

8.2.2 Guided Instruction: More Key Features of Quadratics

Component Overview: *Students receive guided instruction on additional key features for quadratics, focusing on the end behavior, domain, and range. This component ends with students exploring the relationship between the vertex and the x-intercepts.*

8.2.2 Guided Instruction: More Key Features of Quadratics

Two different quadratic equations are shown on the coordinate grid.



The ***end behavior*** of a function refers to the appearance of the function's graph at the "ends," in other words, as the x-values get infinitely smaller or as the x-values get infinitely larger.

Leverage a vocabulary note-taking protocol to introduce the concept of end behavior at this point in the lesson. Consider providing additional examples of what end behavior of different graph types would look like and how each of them has end behavior.

Use the definition of end behavior to complete each statement..

1. For the **red** parabola, as the x-values get infinitely smaller, the ends of the graph of the **red** parabola approach
   - ○ $-\infty$.
   - ○ ∞.

8.2.2 Guided Instruction: More Key Features of Quadratics (continued)

2. For the **red** parabola, as the x-values get infinitely larger, the ends of the graph of the **red** parabola approach
   - ○ $-\infty$.
   - ○ ∞.

3. For the **purple** parabola, as the x-values get infinitely smaller, the ends of the graph of the **purple** parabola approach
   - ○ $-\infty$.
   - ○ ∞.

4. For the **purple** parabola, as the x-values get infinitely larger, the ends of the graph of the **purple** parabola approach
   - ○ $-\infty$.
   - ○ ∞.

The ***domain*** is the complete set of possible values of the input of a function or relation. The domain may vary depending on the context.

5. The equation of the **red** parabola is $y = -2(x + 3)^2 + 2$. Based on the definition of domain, what values can be used as an input for x for the **red** parabola?
   $-\infty < x < \infty$

6. The equation of the **purple** parabola is $y = 2(x - 1)^2 + 2$. Based on the definition of domain, what values can be used as an input for x for the **purple** parabola?
   $-\infty < x < \infty$

Consider an inquiry-based approach during this section.

- *What do you notice that is different about the end behaviors of the red parabola versus the purple parabola?*
- *Why do you think that they would have different end behaviors?*
- *What is the end behavior related to whether the function opens up or down?*

Students may struggle with understanding why the inequality form in questions 5 and 6 uses $<$ or $>$ rather than $\leq$ or $\geq$. Ask students if ∞ can be noted as a value. Have students give the largest value they can think of and then state "I can add 1." This helps students understand the need for $<$ or $>$ rather than $\leq$ or $\geq$.

7. Make a conjecture about the domain of quadratic functions.
Answers will vary. The domain of quadratic functions will be all real numbers. I think that all values on the x-axis would be part of the domain.

The ***range*** is the complete set of possible values of the output of a relation or function.

What are the similarities and differences between domain and range?

8. The equation of the **red** parabola is $y = -2(x+3)^2 + 2$. Based on the definition of range, what values are possible for y for the **red** parabola?
$-\infty < y \leq 2$ *or* $y \leq 2$

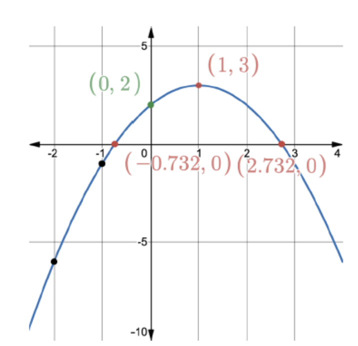
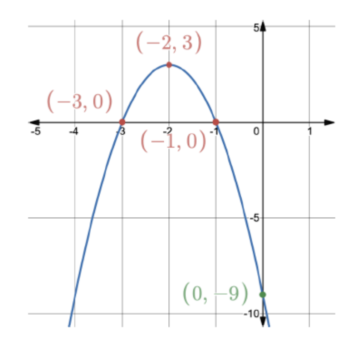
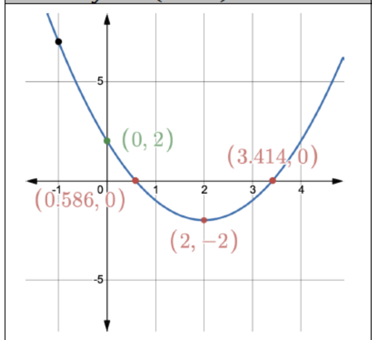
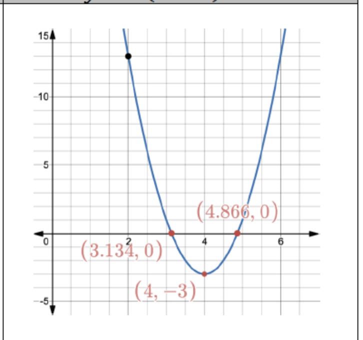
9. The equation of the **purple** parabola is $y = 2(x-1)^2 + 2$. Based on the definition of range, what values are possible for y for the **purple** parabola?
$2 \leq y < \infty$ *or* $y \geq 2$

10. Make a conjecture about the range for all quadratic functions.
Answers will vary. The range for all quadratics will be greater than or equal to or less than or equal to the y-value of the vertex. I think the range for quadratics won't go beyond the vertex of the parabola.

Allow students to share out their conjectures for question 10. Have them explain why they think that their conjecture is accurate. Consider polling the class to see who agrees and disagrees with each conjecture, providing information on why they agree or disagree.

8.2.2 Guided Instruction: More Key Features of Quadratics (continued)

11. Use the given graphs to discuss with your partner your conjecture about domain and range for all quadratics. If necessary, update your conjectures.

<table>
<tr><th>$y = -1(x-1)^2 + 3$</th><th>$y = -3(x+2)^2 + 3$</th></tr>
<tr><td>
</td><td>
</td></tr>
<tr><th>$y = 1(x-2)^2 - 2$</th><th>$y = 4(x-4)^2 - 3$</th></tr>
<tr><td>
</td><td>
</td></tr>
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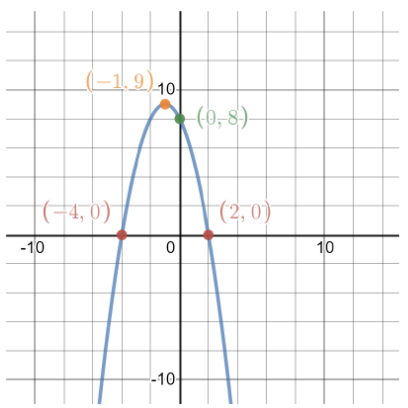
Utilize the Desmos activity that accompanies this lesson in order to display each graph so students can navigate each to explore and manipulate key features.

What do you notice about each of these graphs?

12. Use the graphs to determine if there is a relationship between the equation and whether the parabola opens up or down.
Answers will vary. If there is a negative after the equal sign, the parabola opens downward. If there is no negative after the equal sign, the parabola opens upward.

A table and a graph of the quadratic $y = -1(x+1)^2 + 9$ is shown. The key features of the graph are color coded with the key features in the table. Also, five different forms of the equation of the quadratic are shown with the key features found in each form color coded.

Graph



Key Features

x-intercept	$(-4,0)$
x-intercept	$(2,0)$
y-intercept	$(0,8)$
vertex	$(-1,9)$
axis of symmetry	$x = -1$
increasing	$x < -1$
decreasing	$x > -1$

Key Features

positive	$-4 < x < 2$
negative	$x < -4$ or $x > 2$
domain	R
range	$y \leq 9$
left end behavior	$x \to -\infty,\ y \to -\infty$
right end behavior	$x \to \infty,\ y \to -\infty$
opens	**down**

Equations

$$y = -1(x - (-1))^2 + 9$$

$$y = -1(x+1)^2 + 9$$

$$y = -1(x - (-4))(x - 2)$$

$$y = -1(x+4)(x-2)$$

$$y = -1x^2 - 2x + 8$$

13. How does the fact that the parabola opens down help to determine the end behavior?
Answers will vary. If the parabola opens down, then the end behavior for when x approaches infinity and negative infinity will always be negative infinity.

Consider a brief Turn and Talk before eliciting responses whole group. Clarify any misconceptions using the example below.

Notice the use of colors seen in the key. Consider different colored highlighters or markers when teaching on the master display during instruction. Sometimes the technology distorts the colors. Students will benefit from seeing a color connection between the two representations.

Consider brief synthesizing questions for students to make sense of their work.

- *What is the ______ (key feature)?*
- *How did you determine this key feature? Where does it appear (by representation, equation, graph, or table of values)?*

For a challenge, have students try different quadratic equations to see if their answer to question 13 is consistent with different equation types.

"Prove your answer with at least one example. Provide any counterexamples if you disagree with a partner's answer."

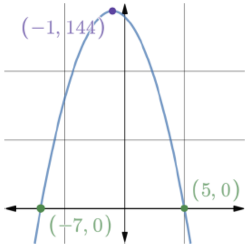
If technology is available, have students use a graphing utility, to verify their answers.

8.2.2 Guided Instruction: More Key Features of Quadratics (continued)

14. Use the information in the table to determine the relationship between the x-intercepts and the vertex.

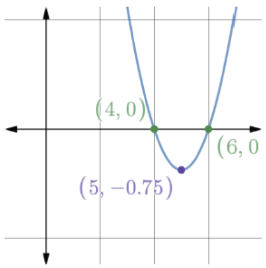
Equations
$y = -4(x+7)(x-5)$ $y = -4(x+1)^2 + 144$
x-intercepts
$(-7,0)$ and $(5,0)$
Vertex
$(-1, 144)$

Graph



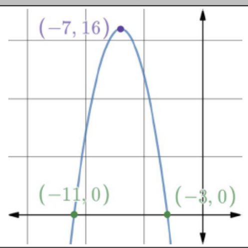
Equations
$y = \frac{3}{4}(x-4)(x-6)$ $y = \frac{3}{4}(x-5)^2 - \frac{3}{4}$
x-intercepts
$(4,0)$ and $(6,0)$
Vertex
$\left(5, -\frac{3}{4}\right)$

Graph



Equations
$y = -(x+11)(x+3)$ $y = -(x+7)^2 + 16$
x-intercepts
$(-11,0)$ and $(-3,0)$
Vertex
$(-7, 16)$

Graph



15. The relationship between the x-intercepts and the vertex is ....
Answers will vary. The x-value of the vertex is the middle of the x-intercepts. To find the y-value, substitute the x-value into the equation.

What is the relationship between the x-intercepts and the vertex? Why do you think that this relationship exists? How does that relationship connect to line of symmetry?

For a challenge, have students try to illustrate the relationship between the vertex and x-intercepts, explaining verbally or on a graph while referencing or highlighting the axis of symmetry.

8.2.3 Collaboration: Using Key Features of Quadratics to Graph

Component Overview: *Students will collaborate to determine the key features of a quadratic. For this activity, review the directions for the students and then release them to work with their partners or in small groups on this activity.*

8.2.3 Collaboration: Using Key Features of Quadratics to Graph

1. Work with your partner to graph $y = -1(x-2)^2 + 3$ to complete the following steps.
   - Determine all of the key features **possible** using the equation and table of values.
   - Use the key features to graph the quadratic.

Equation	Table of Values
$y = -1(x-2)^2 + 3$	see table below

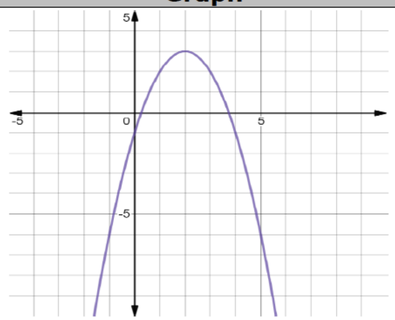
x	y
-1	-6
0	-1
1	2
2	3
3	2
4	-1

Key Features

x-intercept		vertex	$(2,3)$	domain	R
x-intercept		increasing	$x < 2$	range	$y \le 3$
y-intercept	$(0,-1)$	decreasing	$x > 2$	opens	downwards

positive		left end behavior	$x \to -\infty$, $y \to -\infty$
negative		right end behavior	$x \to \infty$, $y \to -\infty$

Graph



MA.K12.MTR.2.1: Multiple Representations
Prompt students to make connections between the various representations of the quadratic, making use of each representation for identifying different key features.

Consider opportunities for building collaboration across groups or pairs. Once time is called, have each group rotate to another group and initial a key feature that they agree with. If they disagree, have them explain their thinking and see if they can get one or the other to change their answer based on the explanation. Repeat until all students have had their key features verified by other groups.

8.2.4 Your Turn!

Component Overview: *Students will work independently to determine the key features of a quadratic. For this activity, release students to complete question 1 on their own, independently. End by bringing students back together to review the correct key features.*

8.2.4 Your Turn!

1. Determine all of the key features possible using the equation of the quadratic $y = 4(x+3)(x-1)$. Then, use the key features to graph the quadratic. If necessary, complete the table of values.

Key Features

x-intercept	$(-3,0)$	vertex	$(-1,-16)$	domain	$\mathbb{R}$
x-intercept	$(1,0)$	increasing	$x > -1$	range	$y \geq -16$
y-intercept	$(0,-12)$	decreasing	$x < -1$	opens	*upwards*

positive	$x < -3$ *or* $x > 1$	left end behavior	$x \to -\infty$, $y \to \infty$
negative	$-3 < x < 1$	right end behavior	$x \to \infty$, $y \to \infty$

Graph

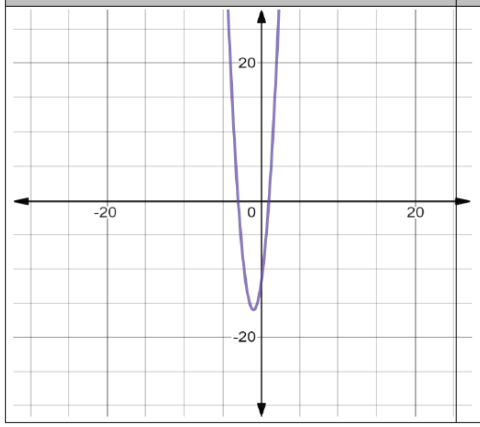


Table of Values

x	y
1	0
0	-12
-1	-16
-2	-12
-3	0
-4	20

When having students share out their answers during the activity closure, consider polling the class in order to see which students got the same answer. Repeat this for each key feature, calling on new students each time. Consider the use of a random name generator tool or popsicle sticks with their names on them.

Consider using this time to provide differentiated support to students who need additional support based on your observations from 8.2.3 Collaboration.

8.2.5 Wrap-Up: Reflection

Component Overview: *Students will complete a reflection on their level of understanding for each key feature they have experienced. This activity is designed to be done independently to provide individual student data on their self-assessment of understanding key features.*

8.2.5 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each key feature.
   Answers will vary.

   I need help. I got it. I could help someone.

x-intercepts	
y-intercept	
vertex	
domain	
range	
opens up/down	
increasing/decreasing	
positive/negative	
end behavior	

Mark-Up Text
Consider prompting students to mark each activity with each of the symbols provided in the Wrap-Up to provide more information for you as the teacher and more opportunity for reflection and synthesis for the student.